Classification: Logistic Regression, ROC, and Naive Bayes

Biostatistics 735/794

September 23, 2025

Logistic Regression for Classification

- Logistic regression models the probability of an event:

$$\pi_i = \Pr(Y_i = 1 | X_i) = \frac{e^{X_i^\top \beta}}{1 + e^{X_i^\top \beta}}$$

- Classification requires converting probabilities into a decision:

$$\hat{Y}_i = \begin{cases} 1 & \pi_i \geq c \\ 0 & \pi_i < c \end{cases}$$

- The **threshold** c is often set at 0.5 but can be adjusted depending on the cost of false positives vs. false negatives.
- Example: In disease screening, a threshold lower than 0.5 may increase sensitivity and reduce missed cases.

Probabilities vs. Predicted Classes

- **ROC curve and AUC:**
  - ROC = Receiver Operating Characteristic curve.
  - Plots *Sensitivity (TPR)* vs. *1-Specificity (FPR)* across all possible thresholds.
  - AUC = area under the ROC curve; interpretable as the probability that a randomly chosen case is ranked above a control.
- **Class-based metrics:**
  - Sensitivity, Specificity, PPV, and NPV summarize performance at a *fixed threshold*.
  - These depend on the prevalence of the outcome and the decision threshold.
- Key distinction: ROC/AUC evaluate the *ranking of probabilities* (π_i's), while Se/Sp/PPV/NPV evaluate *predicted labels* ($\hat{Y}_i$).

Naive Bayes: Key Idea

- Uses Bayes' Theorem:

$$\Pr(G = k | X = x) \propto \Pr(G = k)\, f_k(x)$$

- Challenge: estimating the joint density $f_k(x)$ for high-dimensional data.
- **Naive Bayes assumption:** Features are conditionally independent given the class:

$$f_k(x) = \prod_{j=1}^{p} f_{kj}(x_j)$$

- Result: Simple, fast, and effective even in high-dimensional data (e.g., text classification, genetics, EHR data).

From Posterior to Classification

- After estimating class priors $\pi_k = \Pr(G = k)$ and feature distributions $f_{kj}(x_j)$:

$$\Pr(G = k|X = x) = \frac{\pi_k \prod_{j=1}^{p} f_{kj}(x_j)}{\sum_{\ell} \pi_\ell \prod_{j=1}^{p} f_{\ell j}(x_j)}$$

- For a new subject with features x, we compute the posterior probability for each class k.
- **Classification rule:** Assign x to the class with the largest posterior probability.

$$\hat{G}(x) = \arg\max_k \Pr(G = k|X = x)$$

- *Example:* Suppose posteriors are 0.70 for "disease" and 0.30 for "no disease." The classifier predicts "disease."

From Logistic Regression to GAMs

- Standard logistic regression assumes:

$$\log \frac{\pi(x)}{1-\pi(x)} = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

- Predictors enter the model linearly, which may be too restrictive.
- Example: effect of age on disease risk may be non-linear (e.g., flat at younger ages, steep increase after midlife).
- **Generalized Additive Models (GAMs)** extend logistic regression by allowing smooth, non-linear functions of predictors:

$$\log \frac{\pi(x)}{1-\pi(x)} = \beta_0 + f_1(x_1) + f_2(x_2) + \cdots + f_p(x_p)$$

How GAMs Work

- Each $f_j(\cdot)$ is a **smooth function** (often splines) estimated from the data.
- The model still adds these terms together (hence "additive"), keeping interpretation manageable.
- Smoothness is controlled by a penalty:
  - Too smooth $\Rightarrow$ miss important patterns.
  - Too wiggly $\Rightarrow$ risk of overfitting.
- Example formula in R:

```
gam(y   s(age) + bmi + s(bp), family = binomial)
```

- Output: predicted probability of outcome with flexible, data-driven shapes for age and blood pressure.

Why Use GAMs in Public Health?

- Captures complex, realistic exposure–response relationships.
- Provides smoother and more interpretable fits compared to categorizing continuous variables into bins.
- Example applications:
  - Smoking intensity and lung cancer risk.
  - Air pollution exposure and probability of asthma diagnosis.
  - BMI and probability of diabetes onset.
- Visualizations of $f_j(x)$ give insight into how predictors affect the probability of disease.

Estimation Details

The first component to estimate is the **class priors (π_k).** This can be estimated in two ways:

1. **MLE**

$$\hat{\pi}_k = \frac{N_k}{N}$$

   where N_k is the number of samples in class k.
   - Simple and unbiased (for large N).

2. **Smoothing (Laplace / Dirichlet priors):**

$$\hat{\pi}_k = \frac{N_k + \alpha}{N + \alpha}$$

   - Prevents zero probabilities for rare/unseen features.
   - Slight bias, but lower variance and improved robustness.

Naive Bayes with Mixed Data Types

- The second component to estimate is the **class-conditional feature distributions ($f_{kj}(x_j)$):**

$$f_k(x) = \prod_{j=1}^{p} f_{kj}(x_j)$$

- Different feature types are modeled with different conditional distributions:
  - Binary feature (e.g., smoker yes/no) $\rightarrow$ $f_{kj}(x_j)$ is Bernoulli.
  - Count feature (e.g., number of visits)$\rightarrow$ $f_{kj}(x_j)$ is Multinomial (or Poisson).
  - Continuous feature (e.g., BMI) $\rightarrow$ $f_{kj}(x_j)$ is Gaussian.
- In practice, these can be **combined in one model**.
  - $f_k(x)$ multiplies the Bernoulli likelihood, the Multinomial likelihood, and the Gaussian likelihood together.

Estimating Class-Conditional Distributions

- **Bernoulli Naive Bayes (binary features):**
  - Each feature $X_j \in \{0, 1\}$.
  - For class k, estimate success probability:

$$\hat{p}_{kj} = \frac{\#\{x_j = 1 \text{ in class } k\}}{N_k}$$

  - Likelihood term: $f_{kj}(x_j) = \hat{p}_{kj}^{x_j}(1 - \hat{p}_{kj})^{1-x_j}$.
  - Problem: assigns zero probability if a feature never appears in a class. Smoothing (discussed above) is commonly used.
- Note: Bernoulli Naive Bayes is a procedure when all X are Bernoulli.

Estimating Class-Conditional Distributions

- **Multinomial Naive Bayes (counts):**
  - Each feature $X_j = (x_{j1}, x_{j2}, \ldots, x_{jL})$ is a count (e.g., word counts) or category where $x_{j\ell} = 1$ if in category ℓ.
  - Estimate probabilities by normalized counts:

$$\hat{p}_{kj\ell} = \frac{n_{kj\ell}}{\sum_m n_{kjm}}$$

    where $n_{kj\ell}$ is the total count of feature j category ℓ in class k.
  - Likelihood term: $f_{kj}(x_j) \propto \prod_{\ell=1}^{L} \hat{p}_{kj\ell}^{x_{j\ell}}$.
  - Smoothing (discussed above) is commonly used.
- Note: Multinomial Naive Bayes is a procedure when all X are multinomial.

Estimating Class-Conditional Distributions

- **Continuous features (Gaussian Naive Bayes):**
  - Assume $X_j \sim \mathcal{N}(\mu_{kj}, \sigma^2_{kj})$ in class k.
  - Estimate mean and variance from class-specific data.
- **Continuous features (Nonparametric):**
  - Use nonparametric density estimation methods that don't assume any particular form.

Key message: Bernoulli, Multinomial, and Gaussian are not separate methods—they are **choices for feature distributions** within the same Naive Bayes framework.

Adding a Cost Function

- Standard classifiers minimize 0–1 loss:

$$\text{Loss}(\hat{y}, y) = I(\hat{y} \neq y)$$

- In practice, false positives and false negatives may not be equally costly.
- Incorporate a cost matrix $C(k, \ell)$:

$$\hat{y} = \arg\min_{k} \sum_{\ell} C(k, \ell) \Pr(G = \ell | X = x)$$

- Examples:
  - Cancer screening: missing a true case (false negative) may have much higher cost than a false alarm.
  - Predicting hospital readmission: over-predicting (false positives) may waste resources.
- Cost-sensitive classification aligns model decisions with public health priorities.

Summary

- Logistic regression outputs probabilities; thresholding converts them to decisions.
- ROC/AUC summarizes performance over all thresholds; Se/Sp/PPV/NPV summarizes performance at one threshold.
- Naive Bayes simplifies high-dimensional classification by assuming conditional independence.
- Variants (Bernoulli, Multinomial) adapt to different data types.
- Cost-sensitive classifiers allow tailoring decisions to real-world consequences.